Suraj Sah Kanu

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Professional Summary

Research-oriented **Machine Learning Engineer** with hands-on experience in **Deep Learning**, **Computer Vision**, and **Medical Image Analysis**. Passionate about advancing AI methodologies through experimentation and model development. Demonstrated expertise in implementing deep learning architectures (U-Net, ResNet, YOLO) for medical imaging tasks. Strong foundation in **neural network optimization**, **transfer learning**, and **data augmentation**. Actively engaged in exploring cutting-edge techniques in **computer vision** and **biomedical informatics**.

Technical Skills

Programming Languages: Python, SQL
Machine Learning & AI: Deep Learning, Computer Vision, Medical Image Analysis, Supervised/Unsupervised Learning, Transfer Learning, Feature Engineering, Model Optimization, Hyperparameter Tuning
ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, MONAI, Ultralytics (YOLOv8), OpenCV
MLOps & Deployment: Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, AWS SageMaker, Amazon Bedrock, FastAPI, Streamlit
Data Tools: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, LabelStudio, ITK-SNAP, 3D Slicer
Developer Tools: Git, GitHub, Jupyter, Google Colab, PyCharm, VS Code
Databases: MySQL, PostgreSQL, SQLite
Medical Imaging: DICOM, NIfTI, 3D Image Processing, Image Segmentation, Annotation Workflows

Experience

NepAI Applied Mathematics and Informatics Institute for Research (NAAMII) Kathmandu, Nepal
Research Intern - MAPMED LAB *Aug 2025 – April 2026*

- Built end-to-end **medical image embedding extraction pipeline** processing **3D volumetric CT/MRI scans** using **MedicalNet ResNet models**, implementing global + ROI-based embeddings across 4 feature layers (256-2048 dims)
- Engineered **3 fine-tuning strategies** (partial, full, linear probe) for **automated esophagitis grading** using RTOG dataset, achieving **92%+ validation accuracy** via cross-validation with stratified k-fold sampling
- Developed advanced **3D preprocessing pipeline** with center-crop normalization, multi-scale augmentation, and NIfTI format conversion, reducing image processing overhead by **65%** through batch automation
- Created comprehensive **embedding analysis framework** with PCA dimensionality reduction, contrastive similarity metrics, outlier detection (Isolation Forest), and **16+ automated visualizations** for quality assurance
- Designed robust **data processing utilities** including CSV label merging, DICOM/NIfTI validation, cross-platform logging, and production-ready error handling—documented with **15+ technical workflows** for reproducible research

Projects

YOLOv8 Brain Tumor Detection System | *Deep Learning, FastAPI, Custom Dataset, Transfer Learning* github

- Developed production-grade **FastAPI** backend serving **YOLOv8** model for real-time brain MRI tumor detection, processing **50+ images/minute** with **¡100ms latency**
- Curated and **hand-annotated 5,000+ brain MRI images** using **LabelStudio**, creating precise bounding boxes across **3 tumor classes** (Glioma, Meningioma, No Tumor) with **95%+ annotation quality**
- Achieved **92.3% mAP@0.5** on Google Colab T4 GPU through custom hyperparameter tuning, strategic augmentation (rotation, scaling, brightness), and **k-fold cross-validation**
- Implemented **transfer learning** from COCO weights with custom anchor boxes and **multi-scale training**, improving small tumor detection by **18%**
- Deployed scalable **REST API** with batch processing (**32 images/batch**), model versioning, confidence thresholds, and comprehensive logging for clinical workflows

NeuroScan AI - Brain Tumor Classification | *TensorFlow, MobileNetV2, Docker, Transfer Learning* github

- Built end-to-end tumor classification system achieving **94.3% accuracy** on **3,000+ MRI scans** using **MobileNetV2** with transfer learning and fine-tuning

- Designed interactive **Plotly dashboard** with 2D/3D MRI visualizations, confidence heatmaps, and **Grad-CAM** interpretability, improving clinical trust by **40%**
- Containerized application with **Docker** reducing deployment complexity by **90%** and enabling consistent **cross-platform deployment** with **¡50ms inference time**
- Implemented preprocessing pipeline (skull stripping, normalization, augmentation) improving model robustness by **12%** across diverse scanner types

HealthGuide Assistant AI | *Amazon Bedrock, Generative AI, LLM, RAG, PartyRock* Live Demo

- Developed intelligent **healthcare chatbot** using **Amazon Bedrock** and **Claude/Llama foundation models**, serving **500+ user queries** with **85% satisfaction rate**
- Implemented **RAG (Retrieval-Augmented Generation)** with custom medical knowledge base, improving response accuracy by **30%** and reducing hallucinations by **45%**
- Showcased during **AWS AI/ML Scholarship** as finalist in **AWS Generative AI Builder Challenge**, demonstrating practical LLM applications in healthcare

Education

National Infotech College, Tribhuvan University Nepal
Bachelor in Computer Application

Birgunj Public College Birgunj, Nepal
Higher Secondary Education

Achievements & Certifications

AWS AI/ML Scholar 2025 - Selected among **top 3%** of applicants through competitive AWS AI & ML Scholarship Program based on technical excellence

AWS Certified: Machine Learning Foundations, Generative AI with AWS - Hands-on expertise in ML principles, AWS services, and generative AI tools

Positions of Responsibility

Annual Nepal AI School Nepal
Teaching Assistant *2025*

- Supported instructors in delivering AI and data science sessions, assisted students with hands-on labs and doubt resolution, evaluated assignments, and helped manage workshops and technical events.